# CITS4403 Project: Modelling the Perth Housing Crisis

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## 1 Background

Housing instability currently exists as a critical socioeconomic issue within Australian society. Despite high levels of economic prosperity, escalating housing pressures and persistent homelessness continue to affect Australians [2, 3], placing disproportionate burden on marginalised and lower-income groups, such as young people, migrants and renters [6].

Over the past two decades, housing and rental costs in Australia have increased significantly relative to incomes, with home-ownership rates for all age groups under 65 decreasing, particulary among the younger demographic and lower income households [6]. In addition to these general affordability issues, multiple interest rate increases by the Reserve Bank of Australia (RBA) have heavily increased mortgage repayments, creating more financial stress for those lucky enough to enter the housing market [5].

The purpose of this model is to act as a controlled environment to discover and analyses the factors that drive housing instability in Perth. While the majority of studies regarding the housing market are typically based within an Australian context, our model is specific to the Perth CBD landscape.

## 2 Research aims

### 2.1 Research hypothesis

Our hypothesis for this study is to determine which factors have the greatest impact on home ownership rates and housing instability in Perth. Specifically, how factors such as the recent interest rate hikes, the upfront deposit amounts required, and the general income inequality between agents affect housing accessibility. Ultimately, we plan to use our model to evaluate these variables to discover which factor has the most significant influence on Perth's current housing instability.

### 2.2 Contributions

## 3 Prior work

Data-Driven Modelling of Migration, Income Trends, and Rental Pressures to Design Targeted Interventions Reducing Housing Instability [8]
Modelling housing need in Australia to 2025 [10]
London's Housing Crisis; A System Dynamics Analysis of Long-term Developments: 40 Years into the Past and 40 Years into the Future [7]
HousingABM Japan: An Agent-Based Model of Price and Rent Amplification in the Tokyo Housing Market
https://www.comses.net/codebases/0fbece58-7669-4204-870f-1643e7821fb5/releases/1.0.0/ (not peer reviewed)

*Model that our base model was inspired by:*

General Housing Model [4]

# 4 Model specification

**Classes**

`Dwellings`: Represents the houses in our model. Each `Dwelling` is initialised with a tuple representing a set of co-ordinates (*coord*), which then determines it's *quality* based on its location to hotspots, such as the Perth CBD. It stores the current *occupant* and *owner*, (None when initialised), alongside whether it is for sale or not (*isOnMarket*). Lastly, each `Dwelling` has a rent and sale price, formulated using the *AVERAGE_HOUSE_PRICE* (or *AVERAGE_RENT* if it's being used as a rental), multiplyed by it's quality score, and then further multiplied by a random factor to account for real world variations.

`Household`: Represents the people occupying the houses in our model. They're initialised with an income using a lognormal distribution, with the mean being the *AVERAGE_INCOME* variable, and their cash assets (*money*) are set to double their income, so the household can start buying/renting houses immediately (if applicable). A `Household` can occupy a house (*occupying*), and can own multiple houses and mortgages (*owns* and *mortgages* respectively).

`Bank`: Represents the bank. Each `Bank` is initialised with a base interest rate (*interestRate*), and tracks all the active mortgages (*activeMortgages*) occuring in the model. It also hosts an method to update the interest rate each year to simulate the rate changes, with the interest rate unable to go above 17.5% or below 0.1% based on historical data [9].
`Mortgages`:

## 4.1 Assumptions

Note that these assumptions are not yet actually implemented in the model:
Income: Median household income (weekly): $1,865 (ABS 2021 census so slightly out of date) [1]

# 5 Methodology

# 6 Results

# 7 Discussion

# 8 Conclusion

# 9 Future work

### 9.0.1 Credit statement

*Pudney*: Core + solver backend implementation, benchmarking and survey design and implementation, document preparation. *Grinbergs*: loss backend implementation, documentation preparation. *Daggitt*: Conceptualisation, ITP backend and documentation preparation.

# References


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[3] AIHW: Australia's welfare 2023: data insights. Tech. rep., AIHW, Australian Government (2023)

[4] Applegate, J.M.: General housing model (2020). https://doi.org/10.25937/v8yx-4g98

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